

# Jianyuan Zhong

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## Education

### The Chinese University of Hong Kong

PH.D. CANDIDATE AT THE DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Supervisor: Prof. Qiang Xu

Hong Kong, Hong Kong SAR

Aug 2023 - Present

### University of Rochester

B.S. IN COMPUTER SCIENCE WITH MINORS IN MATHEMATICS AND BUSINESS

Graduated with *Excellence in Undergraduate Research in Computer Science*

- Computer Science GPA: 3.65/4.0.

Rochester, NY

Aug 2016 - May 2020

### Hong Kong University of Science and Technology

EXCHANGE STUDENT

Hong Kong, Hong Kong SAR

May 2017 - Aug 2017

## Research Experience

### Ant Group - InclusionAI

RESEARCH INTERN

Supervisor: Prof. Junbo Zhao

**Research focus:** Agentic post-training for diffusion language models.

- Build end-to-end agentic post-training infrastructure on Ant's intra platform, including task-specific harnesses and sandbox environments, SFT trajectory collection pipelines, and an AReal-compatible distributed on-policy training architecture.
- Develop algorithmic innovations tailored to dLLM characteristics.
- Initially brought up CC, search, and math agentic RL end-to-end.

Hangzhou, China

Mar 2026 - Present

### Foundation Model Department, Huawei Hong Kong Research Center

RESEARCH INTERN

Supervisor: Dr. Zhenguo Li

Topic: Formal reasoning, automatic theorem proving, and diffusion language models.

Hong Kong SAR

Mar 2025 - Mar 2026

### The Chinese University of Hong Kong

RESEARCH ASSISTANT

Supervisor: Prof. Qiang Xu

Hong Kong SAR

Oct 2022 - Jul 2023

### Quebec Artificial Intelligence Institute - Mila

RESEARCH INTERN

Supervisor: Prof. Mirco Ravanelli and Prof. Yoshua Bengio

Montreal, Canada

May 2020 - Jun 2021

### Sixth JSALT - Center of Speech and Language Processing @ JHU

RESEARCH ASSISTANT

Supervisor: Prof. Mirco Ravanelli and Prof. Maurizio Omologo

Topic: Noise-robust unsupervised/self-supervised learning of speech representations

Montreal, Canada

Summer 2019

### ROC-HCI Lab, University of Rochester

RESEARCH ASSISTANT

Supervisor: Prof. Ehsan Hoque

Topic: Multimodal learning for understanding human behavior

Rochester, NY

Sep 2018 - May 2020

## Awards & Scholarships

2020 **Computing Research Association Award for Outstanding Undergraduate Researcher**,  
Honorable Mention

USA

2020 **Excellence in Undergraduate Research in Computer Science**, Honor Degree

Rochester, NY

2020 **Charles and Janet Forbes Entrepreneurial Competition - 2nd Prize**, 2,500 USD

Rochester, NY

2016 - 2020 **Dean's Merit Scholarship**, 40,000 USD

Rochester, NY

2016 - 2020 **Dean's List**,

Rochester, NY

## Selected Publications

Jianyuan Zhong, Kaibo Wang, Ding Ding, Zijin Feng, Haoli Bai, Yang Xiang, Jiacheng Sun, and Qiang Xu. Stabilizing reinforcement learning for diffusion language models, 2026 (**RL Post-training for dLLM**)

Jianyuan Zhong\*, Zeju Li\*, Zhijian Xu, Xiangyu Wen, Kezhi Li, and Qiang Xu. Solve-Detect-Verify: Inference-time scaling with flexible generative verifier. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*

*Linguistics*, 2026. **ACL 2026 (Flexible Generative Verifier)**

Yu Xuejun\*, **Jianyuan Zhong\***, Zijin Feng\*, Pengyi Zhai, Roozbeh Yousefzadeh, Wei Chong Ng, Haoxiong Liu, Ziyi Shou, Jing Xiong, Yudong Zhou, Claudia Beth Ong, Austen Jeremy Sugiarto, Yaoxi Zhang, Wai Ming Tai, Huan Cao, Dongcai Lu, Jiacheng Sun, Qiang Xu, SHEN XIN, and Zhenguo Li. Mathesis: Towards formal theorem proving from natural languages. In *The Thirteenth International Conference on Learning Representations*, 2025. **ICLR 2025 (Formal Theorem Proving; ICLR 2025)**

**Jianyuan Zhong\***, Zeju Li\*, Zhijian Xu, Xiangyu Wen, and Qiang Xu. Dyve: Thinking fast and slow for dynamic process verification. *Accepted to EMNLP 2025*, 2025 (**Dynamic Generative PRM; EMNLP 2025**)

Xiangyu Wen\*, **Jianyuan Zhong\***, Zhijian Xu, and Qiang Xu. Guideline Compliance in Task-Oriented Dialogue: The Chained Prior Approach. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 6750–6776, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics (**LLM Dialogue Systems**)

Zeju Li, **Jianyuan Zhong**, Ziyang Zheng, Xiangyu Wen, Zhijian Xu, Yingying Cheng, Fan Zhang, and Qiang Xu. Compressing chain-of-thought in LLMs via step entropy. *arXiv preprint arXiv:2508.03346*, 2025 (**Efficient LLM Reasoning**)

## Technical Reports

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Mirco Ravanelli, Titouan Parcollet, Peter Plantinga, Aku Rouhe, Samuele Cornell, Loren Lugosch, Cem Subakan, Nauman Dawalatabad, Abdelwahab Heba, **Jianyuan Zhong**, Ju-Chieh Chou, Sung-Lin Yeh, Szu-Wei Fu, Chien-Feng Liao, Elena Rastorgueva, François Grondin, William Aris, Hwidong Na, Yan Gao, Renato De Mori, and Yoshua Bengio. SpeechBrain: A general-purpose speech toolkit. *arXiv preprint arXiv:2106.04624*, 2021 (**Core Contributor; SpeechBrain Toolkit; Technical Report at Mila**)

Qiang Xu, Xiangyu Wen, Changran Xu, Zeju Li, and Jianyuan Zhong. From craft to constitution: A governance-first paradigm for principled agent engineering. *Work in Progress to ICML 2026*, 2025 (**Position Paper at CURE, CUHK**)

(\* denotes co-first authorship.)

## Other Publications

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**Jianyuan Zhong\***, Zhijian Xu\*, Saizhuo Wang, Xiangyu Wen, Jian Guo, and Qiang Xu. DSPO: An end-to-end framework for direct sorted portfolio construction. *arXiv preprint*, 2024 (**Portfolio Optimization; arXiv Preprint**)

Xiangyu Wen, Min Li, Junhua Huang, **Jianyuan Zhong**, Zhijian Xu, Zeju Li, Yongxiang Huang, Mingxuan Yuan, and Qiang Xu. Dependency matters: Enhancing LLM reasoning with explicit knowledge grounding. In *Accepted to NeurIPS 2025 Conference*, 2025 (**LLM Reasoning; Submitted to NeurIPS 2025**)

Zeju Li, Changran XU, Zhengyuan Shi, Zedong Peng, Yi Liu, Yunhao Zhou, Lingfeng Zhou, Chengyu Ma, **Jianyuan Zhong**, Xi Wang, Jieru Zhao, Zhufei Chu, Xiaoyan Yang, and Qiang Xu. DeepCircuitX: Repository-level RTL dataset for code understanding, generation, and multimodal analysis. In *IEEE International Conference on LLM-Aided Design*, 2025. **LAD25**

Ziyang Zheng, Shan Huang, **Jianyuan Zhong**, Zhengyuan Shi, Guohao Dai, Ningyi Xu, and Qiang Xu. DeepGate4: Efficient and effective representation learning for circuit design at scale. In *The Thirteenth International Conference on Learning Representations*, 2025. **ICLR**

Yijun Yang, Ruiyuan Gao, Xiao Yang, **Jianyuan Zhong**, and Qiang Xu. GuardT2I: Defending text-to-image models from adversarial prompts. In *The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024

Zhengyuan Shi, Ziyang Zheng, Sadaf Khan, **Jianyuan Zhong**, Min Li, and Qiang Xu. DeepGate3: Towards scalable circuit representation learning. In *Proceedings of the 59th ACM/IEEE Design Automation Conference (DAC)*, 2024

Zhijian Xu, Yuxuan Bian, **Jianyuan Zhong**, Xiangyu Wen, and Qiang Xu. Beyond trend and periodicity: Guide time series forecasting with textual cues. *arXiv preprint*, 2024

Cem Subakan, Mirco Ravanelli, Samuele Cornell, Mirko Bronzi, and **Jianyuan Zhong**. Attention is all you need in speech separation. In *ICASSP 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2021)*, 2021

Mirco Ravanelli, **Jianyuan Zhong**, Santiago Pascual, Pawel Swietojanski, Joao Monteiro, Jan Trmal, and Yoshua Bengio. Multi-task self-supervised learning for robust speech recognition. In *ICASSP 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 6989–6993, 2020

Md Kamrul Hasan, Wasifur Rahman, AmirAli Bagher Zadeh, **Jianyuan Zhong**, Md Iftekhhar Tanveer, Louis-Philippe Morency, and Mohammed (Ehsan) Hoque. UR-FUNNY: A Multimodal Language Dataset for Understanding Humor. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2046–2056, Hong Kong, China, November 2019. Association for Computational Linguistics

(\* denotes co-first authorship.)

## Teaching

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|             |                                                                                              |                        |
|-------------|----------------------------------------------------------------------------------------------|------------------------|
| 2024        | <b>ENGG 2440: Discrete Mathematics for Engineers</b> , Teaching Assistant                    | <i>Hong Kong S.A.R</i> |
| 2023 - 2024 | <b>AIST1110: Introduction to Computing using Python</b> , Teaching Assistant                 | <i>Hong Kong S.A.R</i> |
| 2023        | <b>GPT-Driven Agent for Mocking Coding Interview</b> , Final Year Project Teaching Assistant | <i>Hong Kong S.A.R</i> |
| 2023        | <b>AI in Finance</b> , Final Year Project Teaching Assistant                                 | <i>Hong Kong S.A.R</i> |
| 2022        | <b>AI in Finance</b> , Final Year Project Teaching Assistant                                 | <i>Hong Kong S.A.R</i> |
| 2019        | <b>CSC240: Data Mining</b> , Teaching Assistant                                              | <i>Rochester, NY</i>   |
| 2019 - 2020 | <b>Computer Science Undergraduate Council</b> , Tutor                                        | <i>Rochester, NY</i>   |